

Final Project Overview

Welcome to the final project for the course on **NoSQL Databases with MongoDB and Cassandra**.

In this project, you will apply the knowledge and practical skills you have gained throughout the course to a real-world—inspired data engineering scenario. The tasks in this hands-on project reflect the responsibilities of a Data Engineer working with modern, heterogeneous data platforms in a consulting environment.

Imagine yourself as a Data Engineer at a data analytics consulting firm while completing this project. Your role is to support data analysts by managing, transforming, and querying data across multiple NoSQL database technologies, ensuring that data is accessible, reliable, and ready for analysis.

This **final project**, which will take you approximately 60 minutes to complete, is divided into two parts:

Exercise 1: Work with semi-structured data using MongoDB.

Exercise 2: Design and query a distributed database using Cassandra.

Here are the individual tasks you will perform in each part:

Exercise 1:

- Task 1: Import movies.json into the MongoDB server into an entertainment database and a movie collection.
- Task 2: MongoDB query to find the year in which the most movies were released.
- Task 3: MongoDB query to find the count of movies released after the year 1999.
- Task 4: MongoDB query to determine the average votes for movies released in 2007.
- Task 5: Export the fields from the movies collection into partial_data.csv.

Exercise 2:

- Task 6: Create a keyspace named entertainment.
- Task 7: Import partial_data.csv into a Cassandra server.
- Task 8: Choose the correct option (CQL query to count the movie table's rows.)
- Task 9: Create the index for the movie table's rating column using CQL.
- Task 10: Choose the correct option(CQL query to count the number of movies that are rated "G.")

While performing the tasks, ensure you carefully follow the provided instructions and document your work by capturing outputs and screenshots as required.

Upon completing this assignment, you will demonstrate your ability to work confidently with multiple NoSQL databases, follow industry-standard workflows, and handle real-world data engineering tasks involving diverse data stores.

Let's get started!